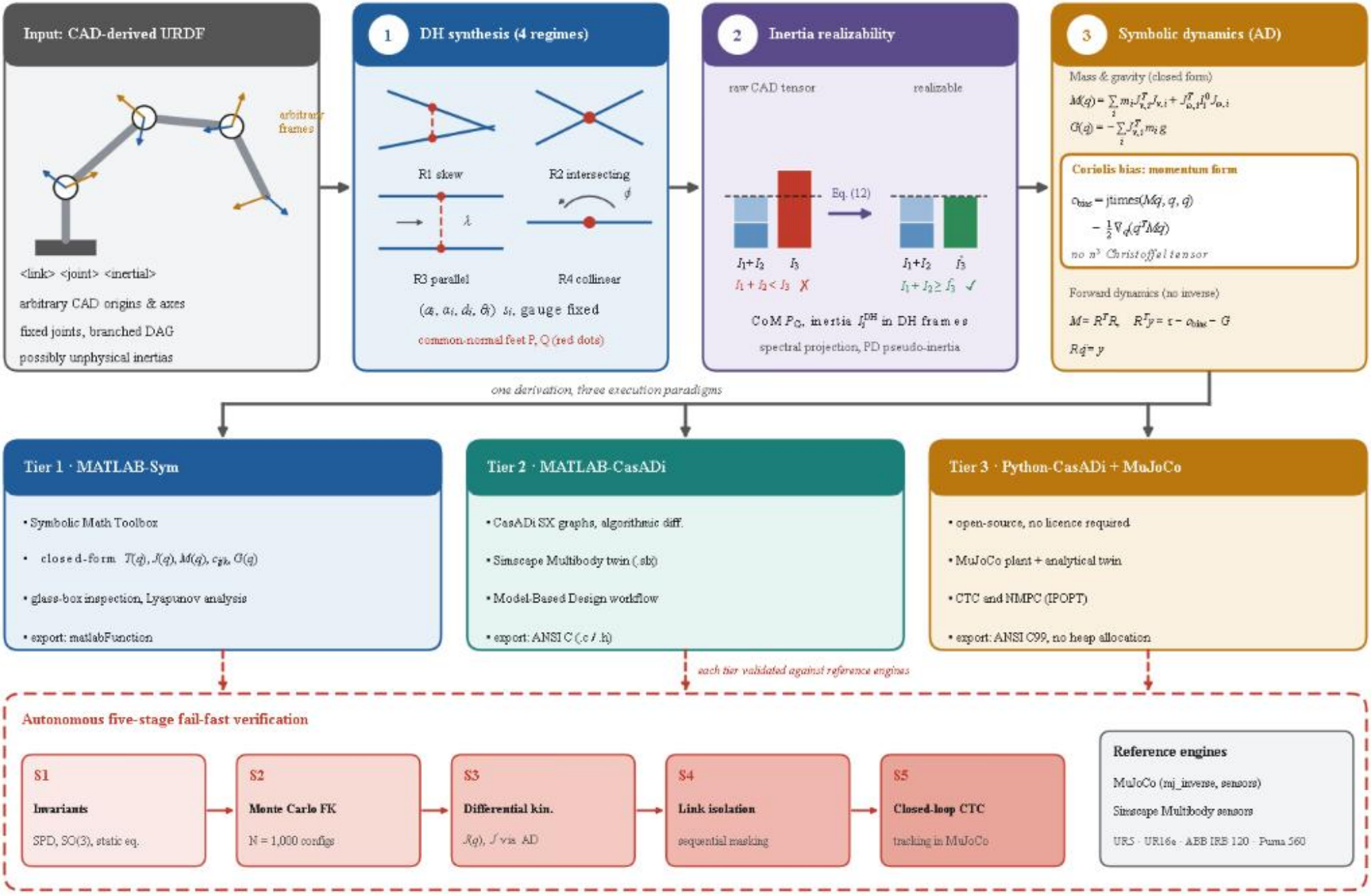

Input: CAD-derived URDF
arbitrary frames
<link> <joint> <inertial>
arbitrary CAD origins & axes
fixed joints, branched DAG
possibly unphysical inertias
1 DH synthesis (4 regimes)
R1 skew
R2 intersecting
R3 parallel
R4 collinear
$(\alpha_i, a_i, d_i, \theta_i)$ s_i, gauge fixed
common-normal feet P, Q (red dots)
2 Inertia realizability
raw CAD tensor
realizable
Eq. (12)
I_1+I_2 I_3
$I_1 + I_2 < I_3$ ✗
I_1+I_2 $\hat{I}_3$
$I_1 + I_2 \geq \hat{I}_3$ ✓
CoM P_Q, inertia I_i^{DH} in DH frames
spectral projection, PD pseudo-inertia
3 Symbolic dynamics (AD)
Mass & gravity (closed form)
$M(q) = \sum_i m_i J_{v,i}^T J_{v,i} + J_{\omega,i}^T I_i^0 J_{\omega,i}$
$G(q) = -\sum_i J_{v,i}^T m_i g$
Coriolis bias: momentum form
$c_{bias} = \text{jtimes}(Mq, q, \dot{q})$
$- \frac{1}{2} \nabla_q (\dot{q}^T M \dot{q})$
no n^3 Christoffel tensor
Forward dynamics (no inverse)
$M = R^T R, \quad R^T y = \tau - c_{bias} - G$
$R\ddot{q} = y$
one derivation, three execution paradigms
Tier 1 · MATLAB-Sym
• Symbolic Math Toolbox
• closed-form $T(q), J(q), M(q), c_{jk}, G(q)$
• glass-box inspection, Lyapunov analysis
• export: matlabFunction
Tier 2 · MATLAB-CasADi
• CasADi SX graphs, algorithmic diff.
• Simscape Multibody twin (.slx)
• Model-Based Design workflow
• export: ANSI C (.c / .h)
Tier 3 · Python-CasADi + MuJoCo
• open-source, no licence required
• MuJoCo plant + analytical twin
• CTC and NMPC (IPOPT)
• export: ANSI C99, no heap allocation
each tier validated against reference engines
Autonomous five-stage fail-fast verification
S1
Invariants
SPD, SO(3), static eq.
S2
Monte Carlo FK
N = 1,000 configs
S3
Differential kin.
$J(q)$, $\dot{J}$ via AD
S4
Link isolation
sequential masking
S5
Closed-loop CTC
tracking in MuJoCo
Reference engines
MuJoCo (mj_inverse, sensors)
Simscape Multibody sensors
URS · UR16e · ABB IRB 120 · Puma 560